



MODULE NAME:	MODULE CODE:
ADVANCED DATABASES	ADDB6311

ASSESSMENT TYPE: PRACTICAL ASSIGNMENT 1 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TIME ALLOWANCE: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. *The research component of the assessment must be typed as well as adequately and correctly referenced.*
2. *Where images are used for reference, they must be included in the submission in a separate document.*
3. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
4. ***Please ensure you attach an originality report to your assignment if required.***
5. ***Make a copy of your assignment before handing it in.***
6. *Consult the marking rubric to familiarise yourself with the assessment criteria to ensure you understand all the submission requirements.*
7. *This is an individual project unless stated otherwise in the brief.*
8. *Keep a clear copy of your project in the form of a digital back-up/a clear photocopy/a photograph.*
9. *Pay careful attention to the submission format and additional requirements as stated in the brief. Make sure your submission is neat and presented professionally.*
10. *All submissions must include a stamped and signed cover sheet, your name and student number.*
11. *Follow all instructions on the brief front page including the additional instructions added below.*

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent** being deducted from the overall percentage awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where **both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Case Study

The following set of relations has been set up for Earl's Kayaking Store specialising in the upgrading of all types of kayaks. At present, the database is small and only includes information about kayaks, customers, upgrades, and kayak upgrades. The relationships between the tables must be derived from the data in each of the tables. The tables and the information required are as follows:

KAYAKS(kayak_id, kayak_type, kayak_model, manufacturer)

CUSTOMER(cust_id, cust_fname, cust_sname, cust_address, cust_contact)

UPGRADES(upgrade_id, upgrade_work, upgrade_date, upgrade_hrs)

KAYAK_UPGRADES (kayak_upgrade_num, kayak_upgrade_date, kayak_upgrade_amt, kayak_id, cust_id, upgrade_id)

Sample Data is shown below:

KAYAKS

KAYAK_ID	KAYAK_TYPE	KAYAK_MODEL	MANUFACTURER
12345	Single Seater	K100	FeelFree
54321	Tandem Seater	J55	Roamer
78945	Fishing Kayak	H9000	Wavesport
98754	Hobie Kayak	A450	Gemini
55311	Canadian Style	L920	Surge

CUSTOMER

CUST_ID	CUST_FNAME	CUST_SNAME	CUST_ADDRESS	CUST_CONTACT
C115	Jeff	Willis	3 Main Road	0821253659
C116	Andre	Watson	13 Cape Road	0769658547
C117	Wallis	Smith	3 Mountain Road	0863256574
C118	Alex	Hanson	8 Circle Road	0762356587

C119	Bob	Bitterhout	15 Main Road	0821235258	
C120	Thando	Zolani	88 Summer Road	0847541254	
C121	Philip	Jackson	3 Long Road	0745556658	
C122	Sarah	Jones	7 Sea Road	0814745745	
UPGRADES					
UPGRADE_ID	UPGRADE_WORK		UPGRADE_DATE	UPGRADE_HRS	
1	Sonar Device		15-JUL-22	5	
2	Padded Seats		18-JUL-22	3	
3	GoPro Camera Mount		19-Jul-22	10	
KAYAK_UPGRADES					
KAYAK_UPGRADE_	KAYAK_UPGRADE_	KAYAK_UPGRADE_	KAYAK_	CUST_	UPGRADE
N	D	A	I	I	_I
UM	ATE	MT	D	D	D
101	27-JUL-19	R 75	98754	C121	3
102	20-JUL-19	R 30	12345	C120	2
103	23-JUL-19	R 75	55311	C119	1
104	17-JUL-19	R 50	54321	C117	1
105	19-JUL-19	R 30	12345	C122	2
You are required to answer the questions that follow.					

Refer to Appendix A at the end of assignment for all Marking Rubrics

Question 1 (Marks: 15)

Management of Earl's Kayaking Store require an Entity Relationship Diagram (ERD) based on the flat files that have been supplied. Your ERD must incorporate all the entities and attributes supplied, as well as establishing the relationships between the entities.

NOTE: It is advised that you use an appropriate drawing tool (e.g. Visio, MS Word, etc.), however you will not be penalised if such tools aren't utilised. If need be, free hand can be used, and a clear and neat screenshot be provided.

Question 2 (Marks: 10)

You will need to create the above tables to complete this **Assignment**. Please create the tables and populate them using SQL Developer or SQL *Plus.

Question 3 (Marks: 10)

Earl's Kayaking Store requires you to create users and grant privileges to use the database. You have been provided with the following details:

Username	Password	Privileges
Tshepo	tmphoabc2023	SELECT ANY TABLE
Mya	mrobertabc2023	INSERT ANY TABLE

Q.3.1	Create the required users and permissions. Note: additional research might be required in line with the Oracle XE version installed.	(8)
Q.3.2	Explain the importance of separation of duties for the different users. Note: include this as a comment at the end of the code.	(2)

Question 4**(Marks: 10)**

Create a SQL query to display the kayak id, customer id, upgrade hours and the kayak upgrade amount. In your query, include the total sales amount by multiplying the upgrade hours by the kayak upgrade amount.

Sample Results

KAYAK_ID	CUST_ID	UPGRADE_HRS	KAYAK_UPGRADE_AMT	TOTAL
98754	C121	10	75	750
12345	C120	3	30	90
55311	C119	5	75	375
54321	C117	5	50	250
12345	C122	3	30	90

Question 5**(Marks: 10)**

Create a SQL query to display the customer's full name, type of kayak, upgrade hours, upgrade work and the kayak upgrade amount.

Sample Results

CUSTOMER	KAYAK_TYPE	UPGRADE_HRS	UPGRADE_WORK	KAYAK_UPGRADE_AMT
Wallis Smith	Tandem Seater	5	Sonar Device	50
Bob Bitterhout	Canadian Style	5	Sonar Device	75
Thando Zolani	Single Seater	3	Padded Seats	30
Philip Jackson	Hobie Kayak	10	GoPro Camera Mount	75
Sarah Jones	Single Seater	3	Padded Seats	30

Question 6**(Marks: 20)**

Complete the following:

- Create a PL/SQL query that will display the customer id, upgrade work done and the upgrade amount. In your query, only display the results where the upgrade amount is greater than R 50.
- Identify the type of cursor used and provide two reasons for its suitability in this query. Provide this information as a comment at the end of your code.
- Provide screenshot of query output.

Sample Results

CUSTOMER ID: C119,

UPGRADE WORK: Sonar Device

UPGRADE AMOUNT: R 75

CUSTOMER ID: C121,

UPGRADE WORK: GoPro Camera Mount

UPGRADE AMOUNT: R 75

Question 7**(Marks: 15)**

Complete the following:

1. Create a PL/SQL query that will display the customer's name, type of kayak, upgrade work and the upgrade date. In your query display the upgrade amount and a discount amount (10%), that will be subtracted off customer 'C121' final bill only.
2. Provide screenshot of query output.

Sample Output

CUSTOMER: Philip Jackson

KAYAK TYPE: Hobie Kayak

UPGRADE WORK: GoPro Camera Mount

UPGRADE DATE: 27-JUL-19

UPGRADE AMT: R 75

DISCOUNT AMT: R 7.5

Question 8**(Marks: 10)**

Create a view called **vwCustUpgrades** that will display the combined customer name, kayak type, upgrade work and the customer contact number. In your query, only display the customers whose address contains the word *'Summer'*.

CUSTOMER	KAYAK_TYPE	UPGRADE_WORK	CONTACT
Thando, Zolani	Single Seater	Padded Seats	0847541254

[TOTAL MARKS: 100]

Appendix A

MODULE NAME:	MODULE CODE:
ADVANCED DATABASES	ADDB6311

STUDENT NAME:	
STUDENT NUMBER:	

Question 1	Levels of Achievement				Feedback
Entity Relationship Diagram	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
Entities	3-4	2	1	0	
	All entities represented in ERD as per flat files supplied.	At least half of the entities represented in ERD as per flat file supplied.	More than half of entities not represented on ERD as per flat file provided.	Not provided	
Attributes	2	1	½	0	
	At least 75% of attributes represented in ERD as per flat files supplied	At least 50% of attributes, represented in ERD as per flat file supplied	At least 10% of attributes represented in ERD as per flat file provided	Not provided	
Keys	4-5	3	1-2	0	

Award ½ per correctly placed PK. Award 1 mark per correctly placed FK	At least 75% of Keys correctly represented in ERD as per flat files supplied	At least 50% of Keys correctly represented in ERD as per flat files supplied	At least 10% of Keys correctly represented in ERD as per flat files supplied	Not provided	
Relationships	3-4	2	1	0	
	At least 75% of relationships correctly identified and implemented in ERD	At least 50% of relationships correctly identified and implemented in ERD	At least 10% of relationships correctly identified and implemented in ERD	Not provided	

Question 2	Levels of Achievement				Feedback
Tables & Data Inserts	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
Database Schema	4-5	3	1-2	0	
	New database schema created; At least 75% of tables created correctly.	New database schema created; At least 50% of tables created correctly.	New database schema created; At least 10% of tables created correctly.	Not provided	
Data values	4-5	3	1-2	0	

	All Data successfully created into each table as per flat files supplied.	Data imported into each table as per flat files supplied; some improvement required	Data imported into some tables as per flat files supplied; major improvements required	Not provided	
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Question 3	Levels of Achievement				Feedback
Users & Permissions	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
Q.3.1	3-4	2	1	0	
	Users and passwords created successfully; Insignificant to no errors in code.	Users and passwords created successfully; minor errors in code requiring minor modifications	Users and passwords created; major errors in code	Not provided	
Q.3.1	3-4	2	1	0	
	All privileges successfully implemented with insignificant to no errors	Privileges implemented with minor modifications required in code	Privileges implemented with major modifications required in code	Not provided 0	
Q.3.2	2	1	½	0	
	SOD well explained in relation to scenario.	SOD theoretically explained.	SOD partially explained with no appropriate reference to scenario.	Not provided	

Question 4 Mark Allocation	Levels of Achievement				Feedback
SELECT QUERY	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
SELECT Statement	3 - 4	2	1	0	
	Correct SELECT statement; Correct columns selected.	Correct SELECT statement used. At least one column not selected.	SELECT statement used; Most columns not selected.	Not provided.	
Tables	3 - 4	2	1	0	
	Correct Tables used; Linked correctly.	Correct Tables used; Links inadequate;	Most Tables not provided; Links incorrect;	Not provided.	
		Minor corrections required.	Major corrections required.		
Query Output	2	1	½	0	
	Correct output achieved; Output Screenshot provided.	Correct output achieved OR Output Screenshot provided.	Correct output not achieved and/or Output Screenshot provided.	Not provided.	

Question 5 Mark Allocation	Levels of Achievement				Feedback
SELECT QUERY	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
SELECT Statement	3 - 4	2	1	0	
	Correct SELECT statement; Correct columns selected.	Correct SELECT statement used. At least one column not selected.	SELECT statement used; Most columns not selected.	Not provided.	
Tables	3 - 4	2	1	0	
	Correct Tables used; Linked correctly.	Correct Tables used; Links inadequate;	Most Tables not provided; Links incorrect;	Not provided.	
		Minor corrections required.	Major corrections required.		
Query Output	2	1	½	0	
	Correct output achieved; Output Screenshot provided.	Correct output achieved OR Output Screenshot provided.	Correct output not achieved and/or Output Screenshot provided.	Not provided.	

Question 6 Mark Allocation	Levels of Achievement				Feedback
	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
Declarations and Variables	3	2	1	0	
	Correct variables declared.	All variables declared; One variable is inadequate/not declared correctly.	At least one variable declared correctly.	Not provided.	
Cursor & SELECT Statement	8 - 10	5 - 7	1 -4	0	
	Correct type of cursor used; Cursor correctly implemented; SELECT statement correctly used; Correct Columns used; Correct tables used & linked correctly.	At least one of the following are inadequate/not implemented; Correct type of cursor used; Cursor correctly implemented; Select statement incorrectly used; Correct Columns used; Correct tables used; Tables linked correctly.	At least two of the following are inadequate/not implemented; Correct type of cursor used; AND the following is inadequate/not implemented Cursor implementation; Use of Select statement.	Not provided.	
Output	2	1	½	0	
	Output Screenshot provided.	Correct output achieved; Screenshot not provided.	Correct output is not achieved; Output Screenshot provided.	Not provided.	
Comments	4 - 5	3	1 -2	0	
	Cursor correctly identified. Correct reasons provided. Information provided as a comment at the end of the code.	Minor changes required	Major changes required	Not provided.	

Question 7 Mark Allocation	Levels of Achievement				Feedback
PL/SQL Query	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
Declarations & Variables	3	2	1	0	
	Correct variables declared.	All variables declared; One variable is inadequate/not declared correctly.	At least one variable declared correctly.	Not provided.	
Cursor & SELECT Statement	8 - 10	5 - 7	1 -4	0	
	Correct type of cursor used; Cursor correctly implemented; SELECT statement correctly used; Correct Columns used; Correct tables used & linked correctly.	At least one of the following are inadequate/not implemented; Correct type of cursor used; Cursor correctly implemented; Select statement incorrectly used; Correct Columns used; Correct tables used; Tables linked correctly.	At least two of the following are inadequate/not implemented; Correct type of cursor used; AND the following is inadequate/not implemented Cursor implementation; Use of Select statement.	Not provided.	
Output	2	1	½	0	
	Correct output achieved;	Correct output achieved; Output	Correct output is not achieved;	Not provided.	
	Output Screenshot provided.	Screenshot not provided.	Output Screenshot provided.		

Question 8 Mark Allocation	Levels of Achievement				Feedback
	Excellent	Good	Developing	Poor	
	Score Ranges Per Level (½ marks possible)				
VIEW Statement	2	1	1	½	
	Correct VIEW statement used and correctly implemented.	VIEW statement used, minor implementation errors.	VIEW statement used, major implementation errors.	Not provided.	
SELECT statements	3-4	2	1	0	
	Correct SELECT statement used. Correct columns	Correct SELECT statement used. At least one required column not selected;	SELECT statement used; Most required columns not selected: Required tables	Not Provided.	
	selected; tables correct	correct Tables mostly used and linked correctly; minor changes required	mostly used, links mostly correct; major changes required		
CODE to view output	2	1	½	0	
	Correct Code to view output is provided.	Code to view output is provided; Minor errors in code.	Code to view output is provided; Major errors in code.	No code provided.	
OUTPUT	2	1	½	0	
	Correct output achieved; Output Screenshot provided.	Correct output achieved; Output Screenshot not provided.	Correct output is not achieved; Output Screenshot provided.	Not provided.	

[TOTAL MARKS: 100]